

Introduction

This document is a supplementary document for the paper entitled *Prioritising Conversion and Comparison of a Fish PBK Model to the Fish Plasma Model for Predicting Plasma Concentrations and Prioritising Data-Poor Pharmaceuticals* and is designed to guide the user on how to use the R-based PBK model. The files required to run the model can be found at <https://github.com/FunkNoodle/FishPBKModel>. While the model runs in R, it is not required for the user to have extensive knowledge of R or scripting. Prior knowledge will help with bug fixing but is not required. The original (equation based) version of the PBK model can be found [here](#).

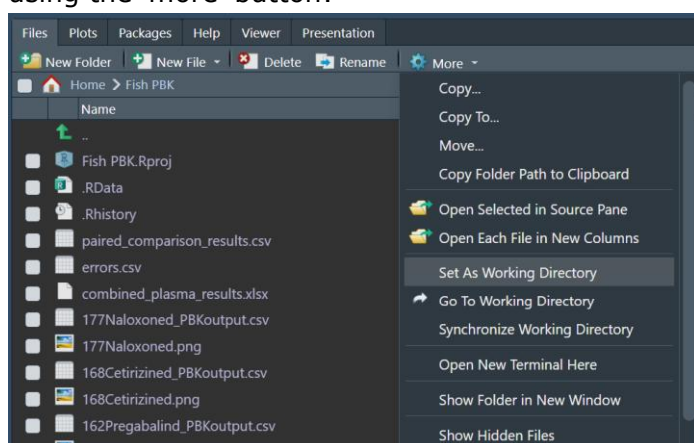
R Installation

If you do not have R installed on your computer, you will need to download both base R, and R studio. Base R is required for R studio to work:

R Studio: <https://posit.co/download/rstudio-desktop/>

Base R: <https://cran.rstudio.com/>

R studio is used to run the model. To get started, place all the model files into a folder, and set that as the working directory when starting a new project in R Studio. Alternatively, you can navigate to the folder using the files tab, and set the working directory manually using the 'more' button:



PBK Model Files

There are four files that are required for the model to run. Two .R script files and a CSV file required for the user input data. These files can be downloaded from <https://github.com/FunkNoodle/FishPBKModel>. Below is a brief overview of what each file does and a screenshot to identify them:

PBK Model.R

This is the master script that contains the PBK model, it may have a version number attached (e.g. *PBK Model v10.R*). You will not need to open this script, but if you are interested in how it runs, it uses the 'Example' data in the PBKinput file (which is actually data for carbamazepine). Do not change anything in this script unless you know what you are doing, otherwise the model will not work as intended.


```

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### Master Script ###

#This runs each .R script in turn

starttime <- Sys.time()
print(starttime)

# Load the necessary library
library(dplyr)

# Read the PBKinput dataframe
PBKinput <- read.csv("PBKinput.csv")

# Read the template script
template_script <- readLines("PBK Model v3.R") # Adjust the path to your template

# Iterate over each name in column A
for (name in PBKinput$Name) {
  # Replace the target_name in the template with the current name
  new_script <- gsub("target_name <- \"Example\"", paste("target_name <- '", name, "'", sep = ""), template_script)

  # Create a new filename for the modified script
  new_filename <- paste0(name, ".R") # Save with the name as part of the filename

  # Write the modified script to a new file
  writeLines(new_script, new_filename)
}

```

PBKinput.csv

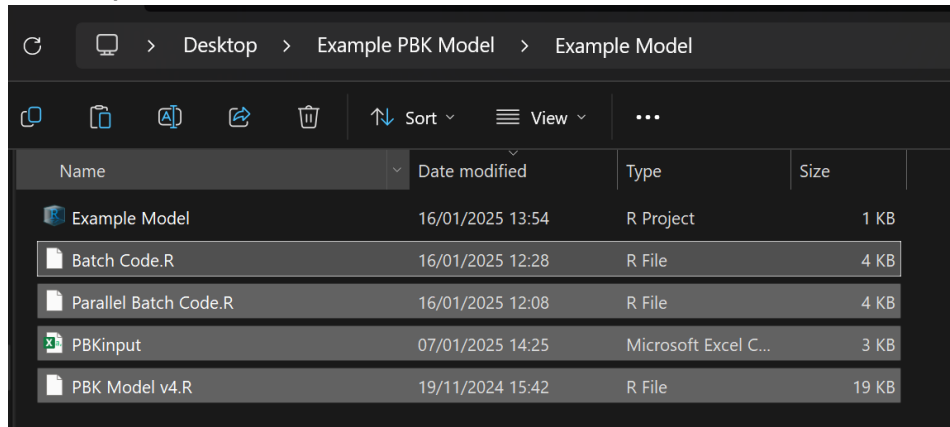
This file is where the user can input their parameters for the model to use. It is recommended to leave the example row as it is for reference. Below is an explanation of each of the column headers:

Column Header	Explanation
Name	The name of the compound or analysis. The user is free to input what they like here so long as there are no duplicates. See figure 3 for examples.
BM	Body Mass: The mass of the fish in grams
Sex	The sex of the fish. 1 for male, 2 for female, 0 for mixed/unknown/other
Temp	Temperature of the environment (surrounding water) in Celsius
pHw	The pH of the surrounding water
pHf	Internal pH of the fish. Leave as 7.4 if you do not have your own data to enter
pKa	The acid dissociation constant of the compound
logKow	The log of the octanol-water partition coefficient (log K _{ow}) of the compound
logKow_ion	The log of the octanol-water partition coefficient (log K _{ow}) of the ionized version of the compound. See the methods section of the main paper to see how to obtain this.
ii	Ionization state: acid = 1, base = -1, neutral = 0, zwitterion = 2 (zwitterion is untested, see SI of the study)
HL	Half-Life: Whole-body primary biotransformation half-life normalised to a 10g fish at 15 °C in days
CL_int	(Optional) Intrinsic clearance, used to calculate hepatic clearance by the model. Set to 0 if you do not want to use this.
UF	(Optional) Unbound Fraction. The model already calculates this, but if you have your own data you can enter it here. Leave as 0 if you do not have any data.
C_water	The concentration of the compound in the surrounding water (ug/ml or mg/l)
Q_ingest	(Optional) Exposure through food (ug/d), set to 0 if there is no ingestion

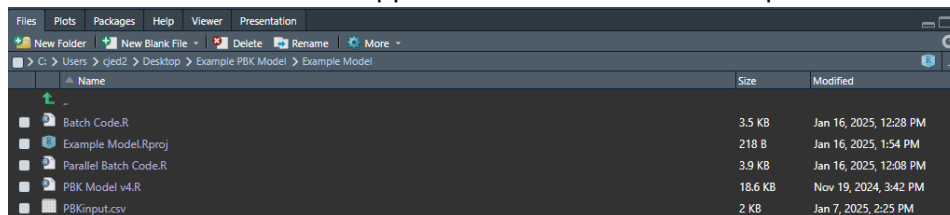
Name	Compound BM	Sex	temp	pHw	pHf	pKa_acid	pKa_base	logKow	logKow_ionii	HL	Cl_int	UF	C_water	Q_ingest		
Example	Carbamaze	10	0	25	7.8	7.4	9.51	0.49	2.43	3.39	0	0.1875	0	0	0.003573	0
P01	Disulfir Disulfiram	10	0	25	7.8	7.4		3.51	3.72	4.16	0	1.144	0	0	0.0367	0
P02	Felbina Felbinac	10	0	25	7.8	7.4	4.46		3.23	0.44	1	2.536	0	0	0.00278	0
P03	Flecain Flecainide	10	0	25	7.8	7.4	10.7	9.11	4.14	1.24	-1	1.491	0	0	0.128757	0
P04	Orphen Orphenadri	10	0	25	7.8	7.4		8.53	3.75	2.25	-1	0.2246	0	0	0.038211	0
P05	Prazosi Prazosin	10	0	25	7.8	7.4		7.18	1.99	-1.65	-1	0.000751	0	0	0.000972	0
P06	Sulpiric Sulpiride	10	0	25	7.8	7.4	9.98	9.12	0.696	-2.01	-1	0.002453	0	0	0.042831	0
P07	Tribeno Tribenoside	10	0	25	7.8	7.4	8.71		4.29	1.83	1	0.05789	0	0	0.0926	0
P08	Blimato Bimatopros	10	0	25	7.8	7.4	14.3	2.13	3.2	0.45	1	0.02037	0	0	1.39E-04	0
P09	Atalure Ataluren	10	0	25	7.8	7.4	4.3	1.53	3.48	0.32	1	6.508	0	0	0.000467	0
P10	Procycl Procyclidin	10	0	25	7.8	7.4	14.2	9.35	4.4	1.17	-1	0.1089	0	0	0.00139	0
P11	Clomet Clomethiaz	10	0	25	7.8	7.4		2.83	2.12	-0.91	-1	0.05445	0	0	0.0608	0
P12	Nimest Nimesulide	10	0	25	7.8	7.4	7.46	6.76	2.62	-0.17	1	0.1116	0	0	0.283453	0
P13	Tolperic Tolperisone	10	0	25	7.8	7.4		8.02	3.65	0.43	-1	0.1798	0	0	0.214153	0
P14	Cilosta Cilostazol	10	0	25	7.8	7.4	9.13	1.83	3.46	0.97	1	0.1207	0	0	0.056201	0
P15	Mephei Mephenox	10	0	25	7.8	7.4	10.5	1.44	1.25	-1.89	1	0.02269	0	0	0.33	0
P16	Heptan Heptamino	10	0	25	7.8	7.4	15.3	10.4	0.829	-1.43	-1	0.1168	0	0	0.154655	0
P17	Indome Indometha	10	0	25	7.8	7.4	6.95		3.88	2.59	1	1.997	0	0	0.0138	0
P18	Clozapi Clozapine	10	0	25	7.8	7.4		8.49	3.49	0.39	-1	0.9172	0	0	9.12E-02	0
P19	Tolbuta Tolbutamid	10	0	25	7.8	7.4	5.3	1.5	2.33	-0.44	1	0.01998	0	0	0.63823	0

Running the model using Batch Code.R

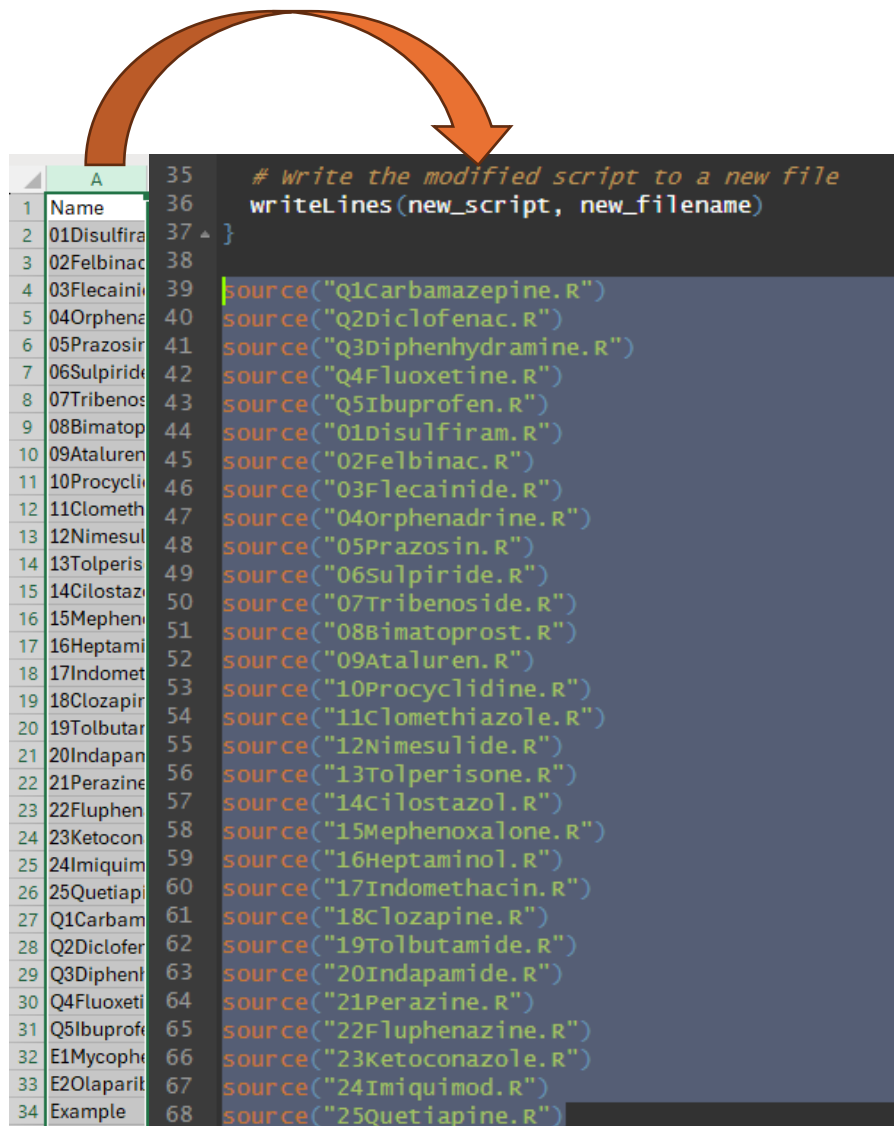
1. File → New Project → New Directory → New Project
 - a. Give your project a name and select an empty folder somewhere on your computer
 - b. Alternatively, you can navigate to the folder using the files tab, and set the working directory manually using the 'more' button (after step 2)
2. Paste in the model files to the new directory (The three .R files and the PBKinput CSV file):



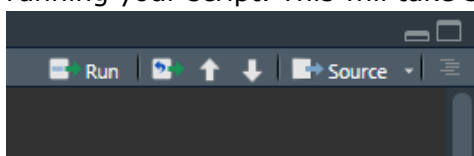
3. These should now appear in the file explorer in R Studio:



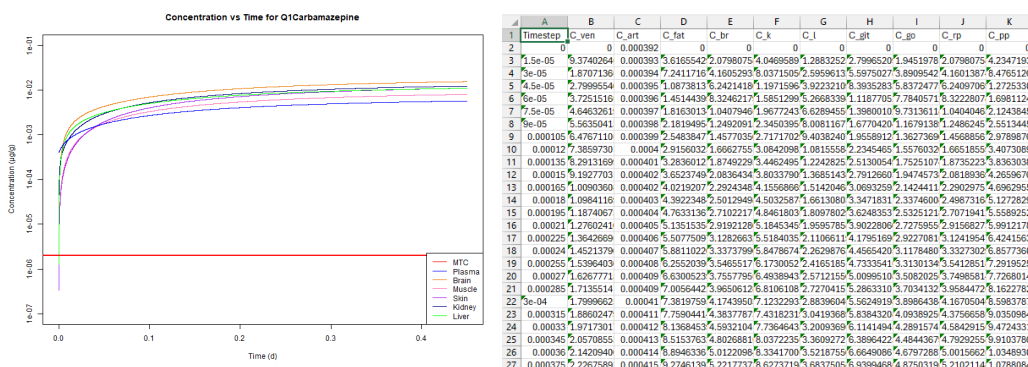
4. Open PBKinput.csv and insert your data. See above for details on each column
5. Open Batch Code.R in R studio. This will open the script. Scroll down to find where the code is going to use the names from the CSV.
 - a. Change the names in green to match the names you put into your CSV file. For example, if you have a row in your Input CSV named "Paracetamol_10mg", then the line in the code should read `source("Paracetamol_10mg.R")`:



6. Press the 'Source' button on the top right of the Batch Code.R script to begin running your script. This will take some time:



7. The model will now go through each compound one by one, and generate a PBKOutputCSV and a graph. You can find both of these either in the folder, or in the file explorer window of R Studio:



- a. The Output CSV contains all of the compartments in the model, and the concentration at each timestep (measured in days). The length of the

timesteps depends on the half-life of the compound ($\text{timestep} = \text{HL}/30,000$).

8. Done!

Once the model has finished running, an ASCII picture of a happy fish will appear in the console, along with the time difference between the start and end of the run, so that you know how long this process takes roughly for future runs.

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>
> print(endtime)
[1] "2025-01-16 14:38:26 CET"
> print(endtime-starttime)
Time difference of 32.87615 mins
>
```

- The code also creates an excel file with a combination of all the final blood plasma results. You may also use this if it is useful to you.

Final Tips

- Timestep (dt) is quite important. If you get strange oscillations, or start to see infinities in your environment, then decrease the timestep by increasing the denominator in either of these equations:

```
80 # dt is a fixed fraction of HL: fine enough to keep forward-Euler stable, and it
81 # shrinks automatically for short-HL compounds (the oscillation fix).
82 # Raise steps_per_HL if a specific compound still oscillates - but note the
83 # results matrix M is preallocated to (steps_per_HL * max_halfives) rows, so
84 # memory scales with the product. 20000 x 60 is about 320 MB; 200000 x 60 would
85 # be about 3 GB and will not allocate.
86 steps_per_HL <- 30000 # timesteps per half-life -> sets dt
87 max_halfives <- 60   # safety cap on run length (a multiple of HL)
```

- If you want your run to be longer, you can also change the maximum run length, which is 60 half-lives by default:

```
80 # dt is a fixed fraction of HL: fine enough to keep forward-Euler stable, and it
81 # shrinks automatically for short-HL compounds (the oscillation fix).
82 # Raise steps_per_HL if a specific compound still oscillates - but note the
83 # results matrix M is preallocated to (steps_per_HL * max_halfives) rows, so
84 # memory scales with the product. 20000 x 60 is about 320 MB; 200000 x 60 would
85 # be about 3 GB and will not allocate.
86 steps_per_HL <- 30000 # timesteps per half-life -> sets dt
87 max_halfives <- 60   # safety cap on run length (a multiple of HL)
```

- You can change the definition of steady-state in the model file by changing the tolerance:

```
442
443 # Steady state = arterial plasma changes by < ss_threshold across a whole
444 # half-life (a fixed TIME window, so the test is independent of how fine dt is).
445 # Compare C_art with its value one half-life (steps_per_HL iterations) ago.
446 ss_threshold <- 1e-3
```

